This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2019.2938302, IEEE Transactions on Automatic Control

Multiple Lyapunov Functions Analysis Approach for Discrete-Time Switched Piecewise-Affine Systems Under Dwell-Time Constraints

Yanzheng Zhu, *Member, IEEE* and Wei Xing Zheng, *Fellow, IEEE*

***Abstract*—In this technical note, the asymptotic stability analysis and state-feedback control design are investigated for a class of discrete-time switched nonlinear systems via the smooth approximation technique. The modal dwell-time switching property is considered to constrain switchings between nonlinear subsystems. A kind of autonomous switchings, i.e., state-partition-dependent switching, is introduced within each approximated piecewise affine (PWA) subsystem. Combining with the state partition information, the novel asymptotic stability conditions with less conservatism are derived for the induced switched PWA systems with dual switching mechanism based on the approach of multiple Lyapunov functions in piecewise quadratic form. Then the design of PWA state-feedback controllers is implemented. The effectiveness of the obtained theoretical results is demonstrated by a numerical example.**

***Index Terms*—Dwell-time switching, multiple Lyapunov functions, piecewise affine approximation, switched nonlinear systems, stabilization.**

I. Introduction

Switched systems consist of a finite number of nonlinear/linear subsystems, together with some switching mechanisms which orchestrate switchings between subsystems. The switched nonlinear systems have attracted a great deal of interest during the past two decades (see, e.g., [1]–[5] and references therein). Considering whether the adopted nonlinear approximation technique is smooth or non-smooth, the corresponding switched nonlinear systems can be approximated as switched piecewise affine (PWA) systems [6], switched polynomial systems [29] and switched linear parameter varying (LPV) systems [7], respectively. Compared to the mushrooming studies on switched LPV systems, the switched PWA systems have received little attention due to the complexity of switching mechanisms, which simultaneously contain the controllable switching (i.e., time-dependent switching) among PWA subsystems and the autonomous switching (i.e., state-partition-dependent switching) within each PWA subsystem. The typical applications on such systems get involved in pneumatic artificial muscle (PAM) [30], urban traffic control [5], switched model predictive control (MPC) [8], etc. In particular, the autonomous switching among different PWA functions of each PWA subsystem causes great difficulties in stability analysis and control synthesis for switched PWA systems, although the results concerning the single PWA system have been extensively reported [9]–[12]. Therefore, there is lack of systematic findings in the area of switched PWA systems either in the discrete-time or continuous-time domain.

This work was supported in part by the National Natural Science Foundation of China under Grants 61973131 and 61603221, in part by the Scientific Research Funds of Huaqiao University under Grant 605-50Y19010, in part by the Australian Research Council under Grant DP120104986, and in part by the NSW Cyber Security Network in Australia under Pilot Grant P00025091.
(Corresponding author: Wei Xing Zheng)

Y. Zhu is with the College of Mechanical Engineering and Automation, Huaqiao University, Xiamen, Fujian 361021, China, and also with the School of Computing, Engineering and Mathematics, Western Sydney University, Sydney, NSW 2751, Australia (e-mail: yanzhengzhu@hqu.edu.cn).

W. X. Zheng is with the School of Computing, Engineering and Mathematics, Western Sydney University, Sydney, NSW 2751, Australia (e-mail: w.zheng@westernsydney.edu.au).

Besides, the existing mathematically intractable results on switched nonlinear systems cannot be extended directly to the case of switched PWA systems with dual switching mechanism, which motivates our current study.

Turning to the controllable switching, the concept of minimal dwell-time was originally born in [13] to deal with the supervisory control of general switched systems. A series of typical applications imply that it is very necessary to have a minimal time interval between two consecutive switching instants, for example, to moderate rise times in circuits [14], to avoid component damage in mechanical systems [15], and to prevent abrupt changes in multi-controller architectures [16]. After that, the notion of minimal dwell-time was recalled in [17] to derive the sufficient conditions of global asymptotic stability of switched linear systems through a novel quadratic Lyapunov function approach. More recently, different analysis approaches have been continuously introduced to reduce conservatism of stability conditions for dwell-time switched linear systems, as well as to increase the design flexibility of dwell-time switching signals, with representative works including the piecewise linear Lyapunov function approach [18], [19], the dwell-time-dependent Lyapunov function approach [20], [21], the homogeneous rational Lyapunov function approach [22] and the clock-dependent Lyapunov function approach [23]. However, when the subsystem is of nonlinear form, even PWA nonlinear form, the corresponding stability analysis and control synthesis have not been adequately explored for discrete-time switched PWA systems under dwell-time switching so far, which provides an opportunity to carry out the current study.

This technical note mainly focuses on the stability analysis and state-feedback control design issues for a class of discrete-time switched nonlinear systems via the smooth approximation technique. The modal dwell-time property is considered as the switching that occurs among nonlinear subsystems, and the PWA approximation, that is smooth, is applied to each nonlinear subsystem. The multiple Lyapunov functions (MLFs) approach is employed here, drawing on the existing results in [24]–[26]. A numerical example is provided to verify the feasibility and effectiveness of the developed theoretical results. The tripartite contributions in this technical note are summarized below. (i) A new class of switched nonlinear approximation models, i.e., the switched PWA models, is presented in this paper. The practical applications on this model have appeared in the areas of urban traffic network control, PAM modeling, and switched MPC, however, the theoretical studies on it are still very insufficient and our current study fills this theoretical gap. (ii) A class of well-known controlled switching signals, i.e., dwell-time switching, is employed to govern the mode variations between PWA subsystems, and an explicitly calculated expression is provided to determine the minimal modal dwell-time for each PWA subsystem. (iii) The MLFs in the piecewise quadratic (PWQ) form are introduced to derive the numerically soluble conditions on asymptotic stability by combining with the state partition information to reduce conservatism, and the corresponding PWA state-feedback controller design is implemented.

The remainder of this technical note is organized as follows.

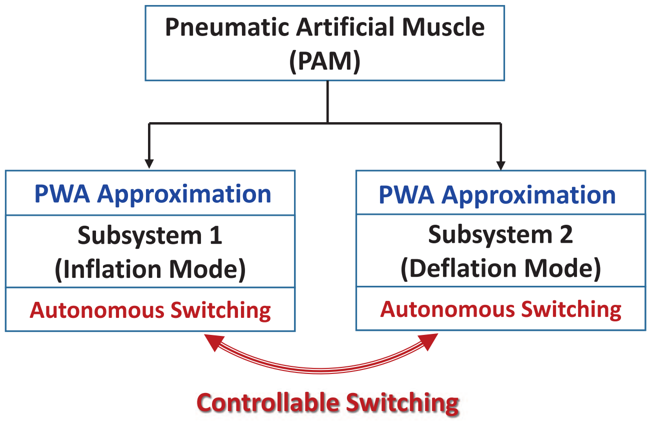


Fig. 1: System setup of PAM based on switched PWA approximation.

A motivating example on the PAM is presented in Section II. In Section III, the problem statement is given. Stability conditions for discrete-time switched PWA systems with dual switching laws are derived based on the MLFs approach in Section IV, and also the PWA controller design is conducted. A numerical example is provided in Section V to demonstrate the effectiveness of the obtained theoretical results. Section VI draws conclusions of this study.

II. Motivating Example: The Pneumatic Artificial Muscle

Pneumatic artificial muscle (PAM), which is frequently used in the field of medical and bio-robotic applications, is of highly nonlinear nature, such that it is rather difficult to carry out the modeling and controller design of it to satisfy the high-performance requirements. A switched piecewise affine (PWA) modeling approach, based on the switching model predictive control (MPC) theory, was developed in [30] to account for alteration of PAM dynamics during inflation and deflation operating states owing to hysteretic phenomena. For each operating mode, the corresponding PAM dynamics is approximated by a family of PWA models at several operating points. The alteration of PAM dynamics during inflation and deflation is scheduled by a controlled switching signal. The system setup of PAM by using a switched PWA model is sketched in Fig. 1.

III. Problem Formulation

Consider a class of discrete-time switched nonlinear system:

$$x_{k+1} = f_{\sigma(k)}(x_k, u_k) \tag{1}$$

where x_k and u_k are the state and the control input of the system, respectively, $\sigma(k)$ is the switching signal which is assumed to be piecewise constant over time, and $f_{\sigma(k)}(\cdot)$ is a nonlinear function. The variable $\sigma(k)$ takes values from a known index set $\mathcal{N} \triangleq \{1, 2, \cdots, N\}$, where $N(>1)$ represents the number of subsystems, and we consider it to be with *dwell time* constraints. As is well known, a function $f \in \digamma \to \mathbb{R}^m$ is piecewise affine (PWA) if there exists a set of polyhedral partitions $\{\digamma_m\}_{m\in\mathcal{S}}$, such that f is affine on each polyhedron $\digamma_m$, where m is the index of state partition. By considering a sufficiently large number of partitions, one can smoothly approximate the nonlinear functions f_i by the PWA form with arbitrary accuracy. For any $\sigma(k) = i \in \mathcal{N}$, the PWA approximation of f_i is expressed in the following form as

$$f_i(x_k, u_k) \cong A_{i,m}x_k + B_{i,m}u_k + a_{i,m}, \tag{2}$$

$\forall x_k \in \digamma_{i,m}$, $m \in \mathcal{S}_i$, $i \in \mathcal{N}$, where $A_{i,m} \in \mathbb{R}^{n\times n}$, $B_{i,m} \in \mathbb{R}^{n\times m}$, $a_{i,m} \in \mathbb{R}^{n\times 1}$ are the system matrices with appropriate dimensions, which can be determined by piecewise linearization for the nonlinear function f_i at m different operating points, and $\{\digamma_{i,m}\}_{m=1}^{S_i}$ represents a polyhedral partition of the set of states $\mathbb{X}_i$ for the i-th subsystem, i.e., each set $\digamma_{i,m}$ is a (not necessarily closed) convex polyhedron such that $\digamma_{i,m} \cap \digamma_{i,n} = \emptyset$, $\forall m \neq n$, and $\cup_{m=1}^{S_i} \digamma_{i,m} = \mathbb{X}_i$, where $m \in \mathcal{S}_i = \{1, \ldots, S_i\}$, $\forall i \in \mathcal{N}$ with S_i being the number of polyhedral partitions induced by PWA approximation of function f_i. The approximation precision for the i-th PWA subsystem mainly depends on the size S_i of state-space partitions $\digamma_{i,m}$. A fine partitioning implies that the obtained PWA approximation has a large number of discrete state values, and the computational complexity on the system analysis and control synthesis will also be enlarged. Each partition is constructed as the intersection of p half spaces and the boundaries are formed by a set of bounding hyperplanes $\mathcal{H}_{i,m}$ defined by

$$\mathcal{H}_{i,m} = \{\, x_k \mid H_{i,m}x_k + h_{i,m} = 0 \,\}\,. \tag{3}$$

Equation (3) defines a collection of half spaces $\Omega_{i,m}$ given by $\Omega_{i,m} = \{\, x_k \mid H_{i,m}x_k + h_{i,m} \geq 0\}$ and $\digamma_{i,m} = \underset{i,m}{\cap} \Omega_{i,m}$. Similar to the notation in [9], the set $\mathcal{S}_i$ is split into $\mathcal{S}_i = \mathcal{S}_{i,0} \cup \mathcal{S}_{i,1}$, where $\mathcal{S}_{i,0}$ denotes the set of indices of the state partitions whose boundary contains the origin for the i-th PWA subsystem while the set $\mathcal{S}_{i,1}$ does not. Without loss of generality, suppose that $a_{i,m} \equiv 0$, $\forall m \in \mathcal{S}_{i,0}$, $i \in \mathcal{N}$.

For the purpose of deriving the main results of the technical note, we consider the following discrete-time switched PWA systems:

$$x_{k+1} = A_{i,m}x_k + B_{i,m}u_k + a_{i,m}, \ \forall x_k \in \digamma_{i,m}, m \in \mathcal{S}_i, i \in \mathcal{N}. \tag{4}$$

In particular, system (4) will degenerate into the switched affine systems as $m \equiv 1$ and $a_{i,m} \neq 0$; into the switched linear systems as $m \equiv 1$ and $a_{i,m} \equiv 0$; into the PWA systems as $i \equiv 1$ and $a_{i,m} \neq 0$; and into the piecewise linear systems as $i \equiv 1$ and $a_{i,m} \equiv 0$.

On the other hand, we assume that the control input u_k is of a PWA state-feedback control law form, i.e.,

$$u_k = K_{i,m}x_k + \kappa_{i,m}, \ \forall x_k \in \digamma_{i,m}, m \in \mathcal{S}_i, i \in \mathcal{N}, \tag{5}$$

where matrices $K_{i,m} \in \mathbb{R}^{m\times n}$ and vectors $\kappa_{i,m} \in \mathbb{R}^{m\times 1}$ are the controller gains to be designed.

Before proceeding further, we present the following results on the stability of discrete-time switched linear systems under dwell-time switching for later use.

Lemma 1 ([18]): Given a known positive scalar $\tau_{d,i} \geq 1$, assume that there exist a set of positive definite symmetric matrices $P_1, P_2, \cdots, P_N$ of compatible dimensions for the switched linear system $x_{k+1} = A_{\sigma(k)}x_k$, such that

$$\begin{cases} A_i^T P_i A_i - P_i < 0, \\ (A_i^{\tau_{d,i}})^T P_j A_i^{\tau_{d,i}} - P_i < 0, \end{cases} \quad \forall i,j \in \mathcal{N}, \ i \neq j. \tag{6}$$

Then the switched linear system $x_{k+1} = A_{\sigma(k)}x_k$ is globally asymptotically stable, and the switching signal $\sigma(k)$ satisfies $t_{k+1} - t_k \geq \tau_{d,i}$, $\forall \sigma(k) = i$ and $k \geq 0$.

Lemma 2 ([13]): Given that $0 < \alpha_i < 1$, $\mu_i > 0$, $i \in \mathcal{N}$, if there exist a series of symmetric matrices $P_i > 0$, such that

$$\begin{cases} A_i^T P_i A_i - P_i < -\alpha_i P_i, \\ P_i \leq \mu_i P_j, \end{cases} \quad \forall i,j \in \mathcal{N}, \ i \neq j,$$

then the system $x_{k+1} = A_{\sigma(k)}x_k$ is globally asymptotically stable with a modal dwell time (MDT) satisfying $\tau_{d,i} \geq -\ln \mu_i / \ln(1-\alpha_i)$.

Remark 1: Lemma 1 provides the best solution for the stability issue of dwell-time switched linear systems. Unfortunately, it cannot be used to deal with the stabilization and robust control issues owing to the existence of the power term on the system matrices A_i, although it is much less conservative than Lemma 2. Lemma 2 is a well-known result about the dwell time incorporated with the MLFs approach, which is also applicable to the case of average dwell-time switching.

For the discrete-time switched PWA system (4) with MDT switching, both the globally uniformly asymptotically stability (GUAS) analysis and the PWA state-feedback controller design are investigated respectively for the first time using the multiple Lyapunov functions approach in this technical note. To present the main results rigorously, two assumptions are introduced in the following, which are essential for the later proofs.

Assumption 1: The origin $x = 0$ is the equilibrium point of system (4).

Note that the stability of the origin will be mainly considered here.

Assumption 2: The dynamics of system (4) is governed by the dynamics of the local model of $\digamma_{i,m}$ when the state of system (4) transits from partition $\digamma_{i,m}$ to partition $\digamma_{i,n} (m \neq n)$ at time constant k.

IV. Main Results

A. Stability Analysis

Given a specific PWA system, we can determine that the current state belongs to which state partition of the current PWA subsystem in real-time by using the reachability analysis approach developed for mixed logical dynamical (MLD) systems [27]. Moreover, when a controlled switching happens, i.e. the state enters into the next PWA subsystem, the index of state partition to which the state belongs could also be confirmed immediately. Hence, regardless of whether the switching is active or not, the corresponding multiple Lyapunov matrices $P_{i,m}$ in the piecewise quadratic (PWQ) form could be searched according to the located partition index of the current state. Based on the above discussion, the GUAS conditions of discrete-time switched PWA systems (4) with MDT switching constraints are given as follows, where it is assumed that $u_k = 0$ to carry out the stability analysis.

Theorem 1: Let $0 < \alpha_i < 1$ be given. Assume that there exist a collection of symmetric matrices $\{\bar{P}_{i,1}, \bar{P}_{i,2}, \cdots, \bar{P}_{i,N}\}$, $\{P_{i,1}, P_{i,2}, \cdots, P_{i,N}\}$ and $\bar{P}_{j,m}$, $P_{j,n}$, $i \neq j$, $Q_{m,n}$, and $U_{i,m}$ such that a series of stability conditions (A1)–(A10), shown at the top of next page, hold, where $\tilde{P}_{i,n} \triangleq \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,n} \begin{bmatrix} I & 0 \end{bmatrix}$, $\tilde{P}_{i,m} \triangleq \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,m} \begin{bmatrix} I & 0 \end{bmatrix}$, $\tilde{A}_{i,m} \triangleq \begin{bmatrix} A_{i,m} & 0 \\ 0 & 1 \end{bmatrix}$, $\bar{S}$ and $\tilde{S}$ can be determined by the reachability analysis method [27], $Q_{m,n}$ and $U_{i,m}$ have nonnegative entries. Then, when $u_k = 0$, the origin of system (4) is GUAS under the switching signal $\sigma(k) = i$ with minimal MDT

$$\tau_{d,i} \geq \tau_{d,i}^* = \frac{\ln(\mu_i)}{-\ln(1-\alpha_i)}, \tag{7}$$

where $\mu_i \triangleq \max_{i \in \mathcal{N}} \{\mu_{0,0}^{[i]}, \mu_{1,1}^{[i]}, \mu_{1,0}^{[i]}, \mu_{0,1}^{[i]}\}$ with $\forall j \in \mathcal{N}$,

$$\mu_{0,0}^{[i]} \triangleq \frac{\lambda_{\max}(P_{j,n})\lambda_{\max}(P_{i,m})}{\lambda_{\min}(P_{i,n})\lambda_{\min}(P_{i,m})},\ \mu_{1,1}^{[i]} \triangleq \frac{\lambda_{\max}(\bar{P}_{j,n})\lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})\lambda_{\min}(\bar{P}_{i,m})},$$

$$\mu_{1,0}^{[i]} \triangleq \frac{\lambda_{\max}(P_{j,n})\lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(P_{i,n})\lambda_{\min}(\bar{P}_{i,m})},\ \mu_{0,1}^{[i]} \triangleq \frac{\lambda_{\max}(\bar{P}_{j,n})\lambda_{\max}(P_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})\lambda_{\min}(P_{i,m})}.$$

Proof: Firstly, we introduce the following multiple Lyapunov functions in the PWQ form as

$$V_i(x_k, m) = \begin{cases} x_k^T P_{i,m} x_k, \ \forall m \in \mathcal{S}_{i,0} \\ \begin{bmatrix} x_k^T & 1 \end{bmatrix} \bar{P}_{i,m} \begin{bmatrix} x_k^T & 1 \end{bmatrix}^T, \ \forall m \in \mathcal{S}_{i,1} \end{cases} \tag{8}$$

where

$$\bar{P}_{i,m} \triangleq \begin{bmatrix} P_{i,m} & p_{i,m} \\ p_{i,m}^T & p_{i,i} \end{bmatrix}, \ \forall m \in \mathcal{S}_{i,1}, i \in \mathcal{N}$$

and $p_{i,m} \in \mathbb{R}^{n\times 1}$ is a vector, $p_{i,i}$ is a scalar as $m \in \mathcal{S}_{i,1}$, and $\bar{P}_{i,m} \triangleq \begin{bmatrix} I_{n\times n} & 0_{n\times 1} \end{bmatrix}^T P_{i,m} \begin{bmatrix} I_{n\times n} & 0_{n\times 1} \end{bmatrix}$ as $m \in \mathcal{S}_{i,0}$. Next, the proof can be divided into two parts: one part is to derive the stability conditions, and the other part is to determine the minimal MDT.

Part I: In the following, borrowed from the analysis approach in [6], four cases are considered to obtain the stability conditions as the post and current states enter into different state partitions, which maybe belong to the same PWA subsystem or not.

Case 1: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$, and $m, n \in \mathcal{S}_{i,0}$, we have

$$\begin{aligned} & \triangle V_i(x_{k+1}, x_k, m) \\ = \ & x_{k+1}^T P_{i,n} x_{k+1} - x_k^T P_{i,m} x_k \\ = \ & x_k^T A_{i,m}^T P_{i,n} A_{i,m} x_k - x_k^T (1-\alpha_i) P_{i,m} x_k - \alpha_i V(x_k, m) \\ = \ & x_k^T M_1 x_k - \alpha_i V(x_k, m), \end{aligned}$$

where $M_1 \triangleq A_{i,m}^T P_{i,n} A_{i,m}^T - (1-\alpha_i) P_{i,m}$. If $M_1 \leq 0$, by using the S-procedure, expression (A3) can be obtained with the aid of state partition information to reduce the conservatism. Then it follows that

$$V_i(x_{k+1}, n) - (1-\alpha_i) V_i(x_k, m) \leq 0. \tag{9}$$

Case 2: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $m, n \in \mathcal{S}_{i,1}$ for the same PWA subsystem i, we have

$$\begin{aligned} & \triangle V_i(\bar{x}_{k+1}, \bar{x}_k, m) \\ = \ & \bar{x}_{k+1}^T \bar{P}_{i,n} \bar{x}_{k+1} - \bar{x}_k^T \bar{P}_{i,m} \bar{x}_k \\ = \ & \bar{x}_k^T \bar{A}_{i,m}^T \bar{P}_{i,n} \bar{A}_{i,m} \bar{x}_k - \bar{x}_k^T (1-\alpha_i) \bar{P}_{i,m} \bar{x}_k - \alpha_i V(\bar{x}_k, m) \\ = \ & \bar{x}_k^T M_2 \bar{x}_k - \alpha_i V(\bar{x}_k, m), \end{aligned}$$

where $M_2 \triangleq \bar{A}_{i,m}^T \bar{P}_{i,n} \bar{A}_{i,m} - (1-\alpha_i) \bar{P}_{i,m}$. If $M_2 \leq 0$, by using the S-procedure, expression (A4) can be derived by considering the state partition information. Then it follows that

$$V_i(\bar{x}_{k+1}, n) - (1-\alpha_i) V_i(\bar{x}_k, m) \leq 0. \tag{10}$$

Case 3: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $m \in \mathcal{S}_{i,1}$, $n \in \mathcal{S}_{i,0}$ for the same PWA subsystem i, we have

$$\begin{aligned} & \triangle V_i(x_{k+1}, \bar{x}_k, m) \\ = \ & x_{k+1}^T P_{i,n} x_{k+1} - \bar{x}_k^T \bar{P}_{i,m} \bar{x}_k \\ = \ & \bar{x}_k^T \left\{ \bar{A}_{i,m}^T \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,n} \begin{bmatrix} I & 0 \end{bmatrix} \bar{A}_{i,m} \right\} \bar{x}_k \\ & - \bar{x}_k^T (1-\alpha_i) \bar{P}_{i,m} \bar{x}_k - \alpha_i V(\bar{x}_k, m) \\ = \ & \bar{x}_k^T M_3 \bar{x}_k - \alpha_i V(\bar{x}_k, m), \end{aligned}$$

where $M_3 \triangleq \bar{A}_{i,m}^T \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,n} \begin{bmatrix} I & 0 \end{bmatrix} \bar{A}_{i,m} - (1-\alpha_i) \bar{P}_{i,m}$. If $M_3 \leq 0$, by using the S-procedure and considering the state partition information, we have

$$\begin{aligned} & \bar{A}_{i,m}^T \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,n} \begin{bmatrix} I & 0 \end{bmatrix} \bar{A}_{i,m} - (1-\alpha_i) \bar{P}_{i,m} \\ & + \bar{E}_{i,m}^T Q_{m,n} \bar{E}_{i,m} \leq 0, \end{aligned}$$

i.e., inequality (A5) holds. Thus, it follows that

$$V_i(x_{k+1}, n) - (1-\alpha_i) V_i(\bar{x}_k, m) \leq 0. \tag{11}$$

Case 4: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $m \in \mathcal{S}_{i,0}$, $n \in \mathcal{S}_{i,1}$ for the same PWA subsystem i, we have

$$\begin{aligned} & \triangle V_i(\bar{x}_{k+1}, x_k, m) \\ = \ & \bar{x}_{k+1}^T \bar{P}_{i,n} \bar{x}_{k+1} - x_k^T P_{i,m} x_k \\ = \ & \bar{x}_k^T \left\{ \begin{bmatrix} A_{i,m} & 0 \\ 0 & 1 \end{bmatrix}^T \bar{P}_{i,n} \begin{bmatrix} A_{i,m} & 0 \\ 0 & 1 \end{bmatrix} \right\} \bar{x}_k - (1-\alpha_i) \\ & \times \bar{x}_k^T \left\{ \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,m} \begin{bmatrix} I & 0 \end{bmatrix} \right\} \bar{x}_k - \alpha_i V(x_k, m) \\ = \ & \bar{x}_k^T M_4 \bar{x}_k - \alpha_i V(x_k, m), \end{aligned}$$

where $M_4 \triangleq \begin{bmatrix} A_{i,m} & 0 \\ 0 & 1 \end{bmatrix}^T \bar{P}_{i,n} \begin{bmatrix} A_{i,m} & 0 \\ 0 & 1 \end{bmatrix} - (1 - \alpha_i) \begin{bmatrix} I & 0 \end{bmatrix}^T P_{i,m} \begin{bmatrix} I & 0 \end{bmatrix}$. If $M_4 \leq 0$, by using the S-procedure

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2019.2938302, IEEE Transactions on Automatic Control

(A1)	$P_{i,m} - E_{i,m}^T U_{i,m} E_{i,m} > 0$	$\forall m \in \mathcal{S}_{i,0}$
(A2)	$\bar{P}_{i,m} - \bar{E}_{i,m}^T U_{i,m} \bar{E}_{i,m} > 0$	$\forall m \in \mathcal{S}_{i,1}$
(A3)	$A_{i,m}^T P_{i,n} A_{i,m} - (1-\alpha_i) P_{i,m} + E_{i,m}^T Q_{m,n} E_{i,m} < 0$	$\forall (m,n) \in \bar{\mathcal{S}} \cap \mathcal{S}_{i,0}$
(A4)	$\bar{A}_{i,m}^T \bar{P}_{i,n} \bar{A}_{i,m} - (1-\alpha_i) \bar{P}_{i,m} + \bar{E}_{i,m}^T Q_{m,n} \bar{E}_{i,m} < 0$	$\forall (m,n) \in \bar{\mathcal{S}} \cap \mathcal{S}_{i,1}$
(A5)	$\bar{A}_{i,m}^T \tilde{P}_{i,n} \bar{A}_{i,m} - (1-\alpha_i) \bar{P}_{i,m} + \bar{E}_{i,m}^T Q_{m,n} \bar{E}_{i,m} < 0$	$\forall (m,n) \in \bar{\mathcal{S}}, m \in \mathcal{S}_{i,1}, n \in \mathcal{S}_{i,0}$
(A6)	$\tilde{A}_{i,m}^T \bar{P}_{i,n} \tilde{A}_{i,m} - (1-\alpha_i) \tilde{P}_{i,m} + \bar{E}_{i,m}^T Q_{m,n} \bar{E}_{i,m} < 0$	$\forall (m,n) \in \bar{\mathcal{S}}, m \in \mathcal{S}_{i,0}, n \in \mathcal{S}_{i,1}$
(A7)	$(A_{i,\cdot}^{\tau_i^*})^T P_{j,n} A_{i,\cdot}^{\tau_i^*} - P_{i,m} < 0$	$\forall (m,n) \in \tilde{\mathcal{S}} \cap \mathcal{S}_{i,0}, i \neq j$
(A8)	$(\bar{A}_{i,\cdot}^{\tau_i^*})^T \bar{P}_{j,n} \bar{A}_{i,\cdot}^{\tau_i^*} - \bar{P}_{i,m} < 0$	$\forall (m,n) \in \tilde{\mathcal{S}} \cap \mathcal{S}_{i,1}, i \neq j$
(A9)	$(\tilde{A}_{i,\cdot}^{\tau_i^*})^T \tilde{P}_{j,n} \bar{A}_{i,\cdot}^{\tau_i^*} - \bar{P}_{i,m} < 0$	$\forall (m,n) \in \tilde{S}, m \in S_{i,1}, n \in S_{j,0}, i \neq j$
(A10)	$(\tilde{A}_{i,\cdot}^{\tau_i^*})^T \bar{P}_{j,n} \tilde{A}_{i,\cdot}^{\tau_i^*} - \tilde{P}_{i,m} < 0$	$\forall (m,n) \in \tilde{S}, m \in S_{i,0}, n \in S_{j,1}, i \neq j$

and making use of the state partition information, it yields that

$$M_4 + \bar{E}_{i,m}^T Q_{m,n} \bar{E}_{i,m} \leq 0,$$

i.e., inequality (A6) holds. Then it follows that

$$V_i(\bar{x}_{k+1}, n) - (1-\alpha_i) V_i(x_k, m) \leq 0. \tag{12}$$

Part II: In this part, the minimal MDT within each PWA subsystem is determined by considering the multiple Lyapunov functions properties. Moreover, the conditions of GUAS can be ensured. The MLFs in (8) satisfy the following constraints:

$$\begin{cases} \lambda_{\min}(P_{i,m}) \left\| x_k \right\|^2 \leq V_i(x_k, m) \leq \lambda_{\max}(P_{i,m}) \left\| x_k \right\|^2, \\ \qquad \text{as } x_k \in \digamma_{i,m}, m \in \mathcal{S}_{i,0}; \\ \lambda_{\min}(\bar{P}_{i,m})(\left\| x_k \right\|^2 + 1) \leq V_i(\bar{x}_k, m) \leq \lambda_{\max}(\bar{P}_{i,m})(\left\| x_k \right\|^2 + 1), \\ \qquad \text{as } x_k \in \digamma_{i,m}, m \in \mathcal{S}_{i,1}. \end{cases} \tag{13}$$

Similar to the analyses of Part I, it can also be partitioned into four cases, when the indices of the current and post states belong to different index sets of state partitions containing the origin or not for any $k \in [k_s, k_{s+1})$.

Case I: $x_{k+1} \in \digamma_{i,n}, x_k \in \digamma_{i,m}$ and $m, n \in \mathcal{S}_{i,0}$.

At the i-th PWA subsystem, one has

$$\begin{aligned} \left\| x_k \right\|^2 &\leq \frac{V_i(x_k, n)}{\lambda_{\min}(P_{i,n})} \leq \frac{(1-\alpha_i)^{\tau_{i,1}} V_i(x_{k-\tau_{i,1}}, m)}{\lambda_{\min}(P_{i,n})} \\ &\leq \frac{(1-\alpha_i)^{\tau_{i,1}+\tau_{i,2}} V_i(x_{k-\tau_{i,1}-\tau_{i,2}}, m)}{\lambda_{\min}(P_{i,n})} \\ &\leq \ldots \leq \frac{(1-\alpha_i)^{k-k_s} V_i(x_{k_s}, m)}{\lambda_{\min}(P_{i,n})} \\ &\leq \frac{(1-\alpha_i)^{k-k_s} \lambda_{\max}(P_{i,m})}{\lambda_{\min}(P_{i,n})} \left\| x_{k_s} \right\|^2. \end{aligned}$$

Specially, assume that $k = k_{s+1}$, $k_{s+1} - k_s = \tau_{d,i}^*$,

$$\left\| x_{k_{s+1}} \right\|^2 \leq \frac{(1-\alpha_i)^{\tau_{d,i}^*} \lambda_{\max}(P_{i,m})}{\lambda_{\min}(P_{i,n})} \left\| x_{k_s} \right\|^2.$$

At the switching time constant, the following inequality needs to be satisfied:

$$V_j(x_{k_{s+1}}, n) \leq V_i(x_{k_s}, m). \tag{14}$$

Then there holds that

$$\begin{aligned} & V_j(x_{k_{s+1}}, n) \\ \leq \; & \lambda_{\max}(P_{j,n}) \left\| x_{k_{s+1}} \right\|^2 \\ \leq \; & \lambda_{\max}(P_{j,n}) \frac{(1-\alpha_i)^{\tau_{d,i}^*} \lambda_{\max}(P_{i,m})}{\lambda_{\min}(P_{i,n})} \left\| x_{k_s} \right\|^2 \\ \leq \; & \lambda_{\max}(P_{j,n}) \frac{(1-\alpha_i)^{\tau_{d,i}^*} \lambda_{\max}(P_{i,m})}{\lambda_{\min}(P_{i,n})} \frac{V_i(x_{k_s}, m)}{\lambda_{\min}(P_{i,m})} \\ = \; & (1-\alpha_i)^{\tau_{d,i}^*} \mu_{0,0}^{[i]} V_i(x_{k_s}, m), \end{aligned}$$

where $\mu_{0,0}^{[i]} \triangleq \frac{\lambda_{\max}(P_{j,n}) \lambda_{\max}(P_{i,m})}{\lambda_{\min}(P_{i,n}) \lambda_{\min}(P_{i,m})}$. If $(1-\alpha_i)^{\tau_{d,i}^*} \mu_{0,0}^{[i]} \leq 1$, one has that inequality (14) holds, i.e.,

$$(A_{i,\cdot}^{\tau_{d,i}^*})^T P_{j,n} (A_{i,\cdot}^{\tau_{d,i}^*}) - P_{i,m} \leq 0, \tag{15}$$

where $\tau_{d,i}^* \geq \frac{\ln(\mu_{0,0}^{[i]})}{-\ln(1-\alpha_i)}$ and $A_{i,\cdot}^{\tau_{d,i}^*} = A_{i,l_0}^{x_{l_0}} A_{i,l_1}^{x_{l_1}} \ldots A_{i,l_k}^{x_{l_k}} \ldots$, which shows that the current model depends on the current located state partition. From (15), it is equivalent to the inequality condition (A7).

Case II: $x_{k+1} \in \digamma_{i,n}, x_k \in \digamma_{i,m}$ and $m, n \in \mathcal{S}_{i,1}$.

At the i-th PWA subsystem, there holds that

$$\begin{aligned} \left\| \bar{x}_k \right\|^2 &\leq \frac{V_i(\bar{x}_k, n)}{\lambda_{\min}(\bar{P}_{i,n})} \leq \frac{(1-\alpha_i)^{\tau_{i,1}} V_i(\bar{x}_{k-\tau_1}, m)}{\lambda_{\min}(\bar{P}_{i,n})} \\ &\leq \frac{(1-\alpha_i)^{\tau_{i,1}+\tau_{i,2}} V_i(\bar{x}_{k-\tau_{i,1}-\tau_{i,2}}, m)}{\lambda_{\min}(\bar{P}_{i,n})} \\ &\leq \ldots \leq \frac{(1-\alpha_i)^{k-k_s} V_i(\bar{x}_{k_s}, m)}{\lambda_{\min}(\bar{P}_{i,n})} \\ &\leq \frac{(1-\alpha_i)^{k-k_s} \lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})} \left\| \bar{x}_{k_s} \right\|^2. \end{aligned}$$

Specially, assume that $k = k_{s+1}$, $k_{s+1} - k_s = \tau_{d,i}^*$,

$$\left\| \bar{x}_{k_{s+1}} \right\|^2 \leq \frac{(1-\alpha_i)^{\tau_{d,i}^*} \lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})} \left\| \bar{x}_{k_s} \right\|^2.$$

At the switching time constant, the following inequality should be satisfied:

$$V_j(\bar{x}_{k_{s+1}}, n) \leq V_i(\bar{x}_{k_s}, m). \tag{16}$$

Then one has that

$$\begin{aligned} & V_j(\bar{x}_{k_{s+1}}, n) \\ \leq \; & \lambda_{\max}(\bar{P}_{j,n}) \left\| \bar{x}_{k_{s+1}} \right\|^2 \\ \leq \; & \lambda_{\max}(\bar{P}_{j,n}) \frac{(1-\alpha_i)^{\tau_{d,i}^*} \lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})} \left\| \bar{x}_{k_s} \right\|^2 \\ \leq \; & \lambda_{\max}(\bar{P}_{j,n}) \frac{(1-\alpha_i)^{\tau_{d,i}^*} \lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})} \frac{V_i(\bar{x}_{k_s}, m)}{\lambda_{\min}(\bar{P}_{i,m})} \\ = \; & (1-\alpha_i)^{\tau_{d,i}^*} \mu_{1,1}^{[i]} V_i(\bar{x}_{k_s}, m), \end{aligned}$$

where $\mu_{1,1}^{[i]} \triangleq \frac{\lambda_{\max}(\bar{P}_{j,n}) \lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(\bar{P}_{i,n}) \lambda_{\min}(\bar{P}_{i,m})}$. If $(1-\alpha_i)^{\tau_{d,i}^*} \mu_{1,1}^{[i]} \leq 1$, one has that inequality (16) holds, i.e.,

$$(A_{i,\cdot}^{\tau_{d,i}^*})^T \bar{P}_{j,n} (A_{i,\cdot}^{\tau_{d,i}^*}) - \bar{P}_{i,m} \leq 0, \tag{17}$$

where $\tau_{d,i}^* \geq \frac{\ln(\mu_{1,1}^{[i]})}{-\ln(1-\alpha_i)}$, and $\bar{A}_{i,\cdot}^{\tau_{d,i}^*} = \bar{A}_{i,l_0}^{x_{l_0}} \bar{A}_{i,l_1}^{x_{l_1}} \ldots \bar{A}_{i,l_k}^{x_{l_k}} \ldots$. Therefore, condition (A8) holds through (17).

Case III: $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $m \in \mathcal{S}_{i,1}$, $n \in \mathcal{S}_{i,0}$.

At the i-th PWA subsystem, one obtains from (13) that

$$\begin{aligned} \left\| x_k \right\|^2 &\leq \frac{V_i(x_k, n)}{\lambda_{\min}(P_{i,n})} \leq \frac{(1-\alpha_i)^{\tau_{i,1}} V_i(\bar{x}_{k-\tau_1}, m)}{\lambda_{\min}(P_{i,n})} \\ &\leq \frac{(1-\alpha_i)^{\tau_{i,1}+\tau_{i,2}} V_i(x_{k-\tau_{i,1}-\tau_{i,2}}, m)}{\lambda_{\min}(P_{i,n})} \\ &\leq \ldots \leq \frac{(1-\alpha_i)^{k-k_s} V_i(\bar{x}_{k_s}, m)}{\lambda_{\min}(P_{i,n})} \\ &\leq \frac{(1-\alpha_i)^{k-k_s} \lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(P_{i,n})} \left\| \bar{x}_{k_s} \right\|^2. \end{aligned}$$

Specially, assume that $k = k_{s+1}$, $k_{s+1} - k_s = \tau^*_{d,i}$,

$$\|x_{k_{s+1}}\|^2 \leq \frac{(1-\alpha_i)^{\tau^*_{d,i}}\lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(P_{i,n})}\|\bar{x}_{k_s}\|^2.$$

At the switching time constant, it is needed that

$$V_j(x_{k_{s+1}}, n) \leq V_i(\bar{x}_{k_s}, m) \tag{18}$$

holds. Then one gets that

$$\begin{aligned}
& V_j(x_{k_{s+1}}, n) \\
\leq\ & \lambda_{\max}(P_{j,n})\|x_{k_{s+1}}\|^2 \\
\leq\ & \lambda_{\max}(P_{j,n})\frac{(1-\alpha_i)^{k-k_s}\lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(P_{i,n})}\|\bar{x}_{k_s}\|^2 \\
\leq\ & \lambda_{\max}(P_{j,n})\frac{(1-\alpha_i)^{k-k_s}\lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(P_{i,n})}\frac{V_i(\bar{x}_{k_s}, m)}{\lambda_{\min}(\bar{P}_{i,m})} \\
=\ & (1-\alpha_i)^{\tau^*_{d,i}}\mu^{[i]}_{1,0}V_i(\bar{x}_{k_s}, m),
\end{aligned}$$

where $\mu^{[i]}_{1,0} \triangleq \frac{\lambda_{\max}(P_{j,n})\lambda_{\max}(\bar{P}_{i,m})}{\lambda_{\min}(P_{i,n})\lambda_{\min}(\bar{P}_{i,m})}$. If $(1-\alpha_i)^{\tau^*_{d,i}}\mu^{[i]}_{1,0} \leq 1$, it follows that inequality (18) holds, i.e.,

$$(\tilde{A}^{\tau^*_{d,i}}_{i,\cdot})^T\tilde{P}_{j,n}(\tilde{A}^{\tau^*_{d,i}}_{i,\cdot}) - \bar{P}_{i,m} \leq 0, \tag{19}$$

where $\tau^*_{d,i} \geq \frac{\ln(\mu^{[i]}_{1,0})}{-\ln(1-\alpha_i)}$, and $\tilde{A}^{\tau^*_{d,i}}_{i,\cdot} = A^{x_{l_0}}_{i,l_0}\bar{A}^{x_{l_1}}_{i,l_1}\ldots A^{x_{l_k}}_{i,l_k}\ldots$. Obviously, condition (A9) is identical to inequality (19).

Case IV: $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $m \in \mathcal{S}_{i,0}$, $n \in \mathcal{S}_{i,1}$.

At the i-th PWA subsystem, by (13), we have

$$\begin{aligned}
\|\bar{x}_k\|^2 \ \leq\ & \frac{V_i(\bar{x}_k, n)}{\lambda_{\min}(\bar{P}_{i,n})} \leq \frac{(1-\alpha_i)^{\tau_{i,1}}V_i(x_{k-\tau_1}, m)}{\lambda_{\min}(\bar{P}_{i,n})} \\
\leq\ & \frac{(1-\alpha_i)^{\tau_{i,1}+\tau_{i,2}}V_i(\bar{x}_{k-\tau_{i,1}-\tau_{i,2}}, m)}{\lambda_{\min}(\bar{P}_{i,n})} \\
\leq\ & \ldots \leq \frac{(1-\alpha_i)^{k-k_s}V_i(x_{k_s}, m)}{\lambda_{\min}(\bar{P}_{i,n})} \\
\leq\ & \frac{(1-\alpha_i)^{k-k_s}\lambda_{\max}(P_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})}\|x_{k_s}\|^2.
\end{aligned}$$

Next, assume that $k = k_{s+1}$, $k_{s+1} - k_s = \tau^*_{d,i}$,

$$\|\bar{x}_{k_{s+1}}\|^2 \leq \frac{(1-\alpha_i)^{\tau^*_{d,i}}\lambda_{\max}(P_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})}\|x_{k_s}\|^2$$

At the switching time, we need

$$V_j(\bar{x}_{k_{s+1}}, n) \leq V_i(x_{k_s}, m). \tag{20}$$

Then it yields that

$$\begin{aligned}
& V_j(\bar{x}_{k_{s+1}}, n) \\
\leq\ & \lambda_{\max}(\bar{P}_{j,n})\|\bar{x}_{k_{s+1}}\|^2 \\
\leq\ & \lambda_{\max}(\bar{P}_{j,n})\frac{(1-\alpha_i)^{k-k_s}\lambda_{\max}(P_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})}\|x_{k_s}\|^2 \\
\leq\ & \lambda_{\max}(\bar{P}_{j,n})\frac{(1-\alpha_i)^{k-k_s}\lambda_{\max}(P_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})}\frac{V_i(x_{k_s}, m)}{\lambda_{\min}(P_{i,m})} \\
=\ & (1-\alpha_i)^{\tau^*_{d,i}}\mu^{[i]}_{0,1}V_i(x_{k_s}, m),
\end{aligned}$$

where $\mu^{[i]}_{0,1} \triangleq \frac{\lambda_{\max}(\bar{P}_{j,n})\lambda_{\max}(P_{i,m})}{\lambda_{\min}(\bar{P}_{i,n})\lambda_{\min}(P_{i,m})}$. If $(1-\alpha_i)^{\tau^*_{d,i}}\mu^{[i]}_{0,1} \leq 1$, then one obtains that inequality (20) holds, i.e.,

$$(\tilde{A}^{\tau^*_{d,i}}_{i,\cdot})^T\bar{P}_{j,n}(\tilde{A}^{\tau^*_{d,i}}_{i,\cdot}) - \tilde{P}_{i,m} \leq 0, \tag{21}$$

where $\tau^*_{d,i} \geq \frac{\ln(\mu^{[i]}_{0,1})}{-\ln(1-\alpha_i)}$, and $\tilde{A}^{\tau^*_{d,i}}_{i,\cdot} = \bar{A}^{x_{l_0}}_{i,l_0}A^{x_{l_1}}_{i,l_1}\ldots\bar{A}^{x_{l_k}}_{i,l_k}\ldots$. Thus, it is straightforward that condition (A10) holds.

Combining *Cases I–IV* together and setting

$$\mu_i \triangleq \max_{i\in\mathcal{N}}\{\mu^{[i]}_{0,0}, \mu^{[i]}_{1,1}, \mu^{[i]}_{1,0}, \mu^{[i]}_{0,1}\},$$

we conclude that $V_{\sigma(k)}(\bar{x}_{k_{s+1}}, n)$ converges to zero as $k \to \infty$. Then the GUAS conditions can be ensured under the MDT switching. Therefore, from the above analyses, it can be deduced that the nominal switched PWA system (4) is GUAS in the Lyapunov sense for all switching sequences satisfying $\tau_{d,i} \geq \tau^*_{d,i}$, which completes the proof. ∎

Remark 2: In Theorem 1, the GUAS conditions are obtained for a class of discrete-time switched PWA systems with dual switching laws. Note that two types of switching signals are considered in (4) as the first attempt. One type is related to the uncontrolled switchings between state partitions of each PWA subsystem, while the other type is the dwell-time switchings between PWA subsystems driven by σ, which is thus controlled. The MLFs in the PWQ form are employed here to derive the stability conditions with less conservatism, in combination with the S-procedure approach. Moreover, the computational complexity grows linearly with respect to the MDT $\tau_{d,i}$. The smaller the MDT is, the lower the computational complexity is, which also verifies that a minimum dwell-time is always expected.

Remark 3: The advantages of the PWQ Lyapunov functions approach include: *i)* The PWQ Lyapunov functions are partition-dependent, which contain the state partition information, and hence relax the restrictions on the Lyapunov function matrices, when compared to the conventional common quadratic Lyapunov functions and even multiple Lyapunov functions; *ii)* The PWQ Lyapunov functions can be obtained by solving a set of linear matrix inequalities (LMIs) via the existing software.

Remark 4: Borrowed from Theorem 1 and the work in Section IV in [5], the stability conditions about the original nonlinear system (1) can be derived by the MLFs approach and the approximation theory. The approximation error can be defined as $\varepsilon_{i,m}(x_k) = f_{i,m}(x_k) - A_{i,m}x_k - a_{i,m}$, $\forall x_k \in \digamma_{i,m}, m \in \mathcal{S}_i, i \in \mathcal{N}$, where the bound on the norm $\|\varepsilon_{i,m}(x_k)\|_2$ of the PWA approximation error can be determined by using the method of Proposition 1 in [5]. The detailed steps on how to obtain this bound are omitted here due to limited space.

B. PWA Controller Design

Denote $\bar{x}_k \triangleq [\ x^T_k \quad 1\ ]^T$. The closed-loop switched PWA systems can be represented as a piecewise linear (PWL) system based on (4) and (5):

$$\begin{cases} \bar{x}_{k+1} = \bar{A}_{i,m}\bar{x}_k + \bar{B}_{i,m}u_k, \ \forall m \in \mathcal{S}_{i,1} \\ x_{k+1} = A_{i,m}x_k + B_{i,m}u_k, \ \forall m \in \mathcal{S}_{i,0} \end{cases} \tag{22}$$

where

$$\bar{A}_{i,m} \triangleq \begin{bmatrix} A_{i,m} & a_{i,m} \\ 0 & 1 \end{bmatrix}, \quad \bar{B}_{i,m} \triangleq [\ B^T_{i,m} \quad 0\ ]^T.$$

In general, the PWA control law (5) can be rewritten as

$$u_k = \begin{cases} K_{i,m}x_k + \kappa_{i,m}, & \forall m \in \mathcal{S}_{i,1} \\ K_{i,m}x_k, & \forall m \in \mathcal{S}_{i,0} \end{cases} \tag{23}$$

Assuming $\bar{K}_{i,m} \triangleq [\ K_{i,m} \quad \kappa_{i,m}\ ]$, expression (22) is equivalent to

$$\begin{cases} \bar{x}_{k+1} = (\bar{A}_{i,m} + \bar{B}_{i,m}\bar{K}_{i,m})\bar{x}_k, \ \forall m \in \mathcal{S}_{i,1} \\ x_{k+1} = (A_{i,m} + B_{i,m}K_{i,m})x_k, \ \forall m \in \mathcal{S}_{i,0} \end{cases} \tag{24}$$

which satisfies $\kappa_{i,m} \equiv 0$ and $a_{i,m} \equiv 0$ as $m \in \mathcal{S}_{i,0}$.

In view of Lemma 2, the state-feedback stabilization controller design is developed using the MLFs approach for closed-loop switched PWA systems (4) in the presence of dual switching laws.

Theorem 2: Suppose that for any given decay rate $0 < \alpha_i < 1$, $\mu_i > 0$, $\varepsilon < 10^{-5}$, there exist symmetric positive definite matrices $\bar{Q}_{i,m}$, $Q_{i,m}$, and matrices $\bar{G}_{i,m}$, $G_{i,m}$, $\bar{Y}_{i,m}$, $Y_{i,m}$, where m is the index of the state partition for the i-th PWA subsystem, and $i \in \mathcal{N}$ is the index of a PWA subsystem, such that the LMIs (B1)–(B8), shown at the top of next page, hold, where $\bar{\mathcal{S}}$ and $\tilde{\mathcal{S}}$ can be determined by the reachability analysis method for mixed logical dynamical (MLD) systems [27]. Then the considered closed-loop switched PWA system (24) is GUAS, which can be stabilized by

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2019.2938302, IEEE Transactions on Automatic Control

(B1)	$\begin{bmatrix} -Q_{i,n} & A_{i,m}G_{i,m}+B_{i,m}Y_{i,m} \\ * & (1-\alpha_i)(Q_{i,m}-G_{i,m}^T-G_{i,m}) \end{bmatrix} \leq 0$	$\forall (m,n) \in \bar{\mathcal{S}} \cap \mathcal{S}_{i,0}$
(B2)	$Q_{i,m} \leq \mu_i Q_{j,n}$	$\forall (m,n) \in \tilde{\mathcal{S}}, m \in \mathcal{S}_{i,0}, n \in \mathcal{S}_{j,0}, i \neq j$
(B3)	$\begin{bmatrix} -\bar{Q}_{i,n} & \bar{A}_{i,m}\bar{G}_{i,m}+\bar{B}_{i,m}\bar{Y}_{i,m} \\ * & (1-\alpha_i)(\bar{Q}_{i,m}-\bar{G}_{i,m}^T-\bar{G}_{i,m}) \end{bmatrix} \leq 0$	$\forall (m,n) \in \bar{\mathcal{S}} \cap S_{i,1}$
(B4)	$\bar{Q}_{i,m} \leq \mu_i \bar{Q}_{j,n}$	$\forall (m,n) \in \tilde{\mathcal{S}}, m \in \mathcal{S}_{i,1}, n \in \mathcal{S}_{j,1}, i \neq j$
(B5)	$\begin{bmatrix} -\tilde{Q}_{i,n} & \bar{A}_{i,m}\bar{G}_{i,m}+\bar{B}_{i,m}\bar{Y}_{i,m} \\ * & (1-\alpha_i)(\bar{Q}_{i,m}-\bar{G}_{i,m}^T-\bar{G}_{i,m}) \end{bmatrix} \leq 0$	$\forall (m,n) \in \bar{\mathcal{S}}, m \in \mathcal{S}_{i,1}, n \in \mathcal{S}_{i,0}$
(B6)	$\bar{Q}_{i,m} \leq \mu_i \tilde{Q}_{j,n}$	$\forall (m,n) \in \tilde{\mathcal{S}}, m \in \mathcal{S}_{i,1}, n \in \mathcal{S}_{j,0}, i \neq j$
(B7)	$\begin{bmatrix} -\bar{Q}_{i,n} & \bar{A}_{i,m}\tilde{G}_{i,m}+\bar{B}_{i,m}\tilde{Y}_{i,m} \\ * & (1-\alpha_i)(\tilde{Q}_{i,m}-\tilde{G}_{i,m}^T-\tilde{G}_{i,m}) \end{bmatrix} \leq 0$	$\forall (m,n) \in \bar{\mathcal{S}}, m \in \mathcal{S}_{i,0}, n \in \mathcal{S}_{i,1}$
(B8)	$\tilde{Q}_{i,m} \leq \mu_i \bar{Q}_{j,n}$	$\forall (m,n) \in \tilde{\mathcal{S}}, m \in \mathcal{S}_{i,0}, n \in \mathcal{S}_{j,1}, i \neq j$

the PWA state-feedback controller (23) with $\bar{K}_{i,m} = \bar{Y}_{i,m}\bar{G}_{i,m}^{-1}$, $K_{i,m} = Y_{i,m}G_{i,m}^{-1}$ for arbitrary switching signals σ with minimal MDT $\tau_{d,i} \geq \tau_{d,i}^* = -\frac{\ln \mu_i}{\ln(1-\alpha_i)}$.

Proof: For the stability analysis, it should be ensured that $V_{\sigma(k)}(x_k, m)$ is positive-definite in a neighborhood of the origin. Then we calculate the difference of the Lyapunov function candidate (8) when $m \in \mathcal{S}_{i,0}$ or $\mathcal{S}_{i,1}$. Similar to the proof steps of Theorem 1, the same four cases are considered in the following.

Case I: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $\forall (m,n) \in \bar{\mathcal{S}} \cap \mathcal{S}_{i,0}$ for the same PWA subsystem i, we have

$$
\begin{aligned}
&\Delta V_i(x_{k+1}, x_k, m) \\
=\ & x_{k+1}^T P_{i,n} x_{k+1} - x_k^T P_{i,m} x_k \\
=\ & x_k^T[(A_{i,m}+B_{i,m}K_{i,m})^T P_{i,n}(A_{i,m}+B_{i,m}K_{i,m})]x_k \\
& -x_k^T(1-\alpha_i)P_{i,m}x_k - \alpha_i V(x_k, m) \\
=\ & x_k^T M_1 x_k - \alpha_i V(x_k, m),
\end{aligned}
$$

where $M_1 \triangleq -(1-\alpha_i)P_{i,m} + (A_{i,m}+B_{i,m}K_{i,m})^T P_{i,n}(A_{i,m} + B_{i,m}K_{i,m})$. If $M_1 \leq 0$, it follows that inequality (9) holds. Then one has

$$
\begin{bmatrix} -P_{i,n} & P_{i,n}(A_{i,m}+B_{i,m}K_{i,m}) \\ * & -(1-\alpha_i)P_{i,m} \end{bmatrix} \leq 0. \tag{25}
$$

Denote that $Q_{i,m} \triangleq P_{i,m}^{-1}$, and $G_{i,m}$ are certain nonsingular matrices of suitable dimensions. Taking the congruence transformation for (25) via $\mathrm{diag}\{Q_{i,n}, G_{i,m}\}$ and its transpose yields that

$$
\begin{bmatrix} -Q_{i,n} & (A_{i,m}+B_{i,m}K_{i,m})G_{i,m} \\ * & -(1-\alpha_i)G_{i,m}^T P_{i,m} G_{i,m} \end{bmatrix} \leq 0. \tag{26}
$$

By using the slack matrix inequality manipulation technique as proposed in [28], the inequality $(Q_{i,m}-G_{i,m})^T P_{i,m}(Q_{i,m}-G_{i,m}) \geq 0$ holds for an arbitrary matrix $G_{i,m}$. Thus, it follows that $Q_{i,m} - G_{i,m} - G_{i,m}^T \geq -G_{i,m}^T P_{i,m} G_{i,m}$. So if inequality (B1) holds, it means that (26) holds, where $Y_{i,m} \triangleq K_{i,m}Q_{i,m}$. Furthermore, at the switching time instant, that is, $x_{k+1} \in \digamma_{j,n}$, $x_k \in \digamma_{i,m}$, $\forall (m,n) \in \tilde{\mathcal{S}}$, $m \in \mathcal{S}_{i,0}$, $n \in \mathcal{S}_{j,0}$, and $i \neq j$, the following inequalities need to be assured according to Lemma 2:

$$
P_{j,n} \leq \mu_i P_{i,m},\ \ i \neq j. \tag{27}
$$

Considering that $P_{i,m} \triangleq Q_{i,m}^{-1}$ and $P_{j,n} \triangleq Q_{j,n}^{-1}$, inequality (27) is equivalent to

$$
Q_{i,m} \leq \mu_i Q_{j,n},\ \ i \neq j, \tag{28}
$$

i.e., inequality (B2) holds.

Case II: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $\forall (m,n) \in \bar{\mathcal{S}} \cap \mathcal{S}_{i,1}$ for the same PWA subsystem i, one has

$$
\begin{aligned}
&\Delta V_i(\bar{x}_{k+1}, \bar{x}_k, m) \\
=\ & \bar{x}_{k+1}^T \bar{P}_{i,n} \bar{x}_{k+1} - \bar{x}_k^T \bar{P}_{i,m} \bar{x}_k \\
=\ & \bar{x}_k^T \hat{A}_{i,m}^T \bar{P}_{i,n} \hat{A}_{i,m} \bar{x}_k - \bar{x}_k^T(1-\alpha_i)\bar{P}_{i,m}\bar{x}_k - \alpha_i V(\bar{x}_k, m) \\
=\ & \bar{x}_k^T M_2 \bar{x}_k - \alpha_i V(\bar{x}_k, m),
\end{aligned}
$$

where $\hat{A}_{i,m} \triangleq \bar{A}_{i,m} + \bar{B}_{i,m}\bar{K}_{i,m}$, and $M_2 \triangleq (\bar{A}_{i,m} + \bar{B}_{i,m}\bar{K}_{i,m})^T \bar{P}_{i,n}(\bar{A}_{i,m} + \bar{B}_{i,m}\bar{K}_{i,m}) - (1-\alpha_i)\bar{P}_{i,m}$. If $M_2 \leq 0$, then it follows that inequality (10) holds. The following LMIs can be obtained according to the derivation steps of *Case I*:

$$
\begin{bmatrix} -\bar{Q}_{i,n} & \bar{A}_{i,m}\bar{G}_{i,m}+\bar{B}_{i,m}\bar{Y}_{i,m} \\ * & (1-\alpha_i)(\bar{Q}_{i,m}-\bar{G}_{i,m}^T-\bar{G}_{i,m}) \end{bmatrix} \leq 0, \tag{29}
$$

that is, inequality (B3) holds, where $\bar{G}_{i,m}$ are nonsingular matrices of suitable dimensions. Moreover, at the switching time instant, that is, $x_{k+1} \in \digamma_{j,n}$, $x_k \in \digamma_{i,m}$, $\forall (m,n) \in \tilde{\mathcal{S}}$, $m \in \mathcal{S}_{i,1}$, $n \in \mathcal{S}_{j,1}$, and $i \neq j$, the following inequalities need to be assured:

$$
\bar{P}_{j,n} \leq \mu_i \bar{P}_{i,m},\ i \neq j
$$

based on the conditions of Lemma 2. Then one obtains

$$
\bar{Q}_{i,m} \leq \mu_i \bar{Q}_{j,n},\ i \neq j, \tag{30}
$$

which is equivalent to inequality (B4), where $\bar{Q}_{j,n} \triangleq (\bar{P}_{j,n})^{-1}$ and $\bar{Q}_{i,m} \triangleq (\bar{P}_{i,m})^{-1}$.

Case III: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $\forall (m,n) \in \bar{\mathcal{S}}$, $m \in \mathcal{S}_{i,1}$, $n \in \mathcal{S}_{i,0}$ for the same PWA subsystem i, we have

$$
\begin{aligned}
&\Delta V_i(x_{k+1}, \bar{x}_k, m) \\
=\ & x_{k+1}^T P_{i,n} x_{k+1} - \bar{x}_k^T \bar{P}_{i,m} \bar{x}_k \\
=\ & \bar{x}_k^T(\bar{A}_{i,m}+\bar{B}_{i,m}\bar{K}_{i,m})^T \begin{bmatrix} P_{i,n} & 0_{n\times 1} \\ * & 0 \end{bmatrix} \\
& \times(\bar{A}_{i,m}+\bar{B}_{i,m}\bar{K}_{i,m})\bar{x}_k - \bar{x}_k^T(1-\alpha_i)\bar{P}_{i,m}\bar{x}_k \\
& -\alpha_i V(\bar{x}_k, m) \\
=\ & \bar{x}_k^T M_3 \bar{x}_k - \alpha_i V(\bar{x}_k, m),
\end{aligned}
$$

where $M_3 \triangleq (\bar{A}_{i,m}+\bar{B}_{i,m}\bar{K}_{i,m})^T \bar{P}_{i,n}(\bar{A}_{i,m}+\bar{B}_{i,m}\bar{K}_{i,m}) - (1-\alpha_i)\bar{P}_{i,m}$, and $\bar{P}_{i,n}$ is irreversible. If $M_3 \leq 0$, then it follows that inequality (11) holds. From $M_3 \leq 0$, there exists a scalar $\varepsilon < 10^{-5}$, such that

$$
\begin{bmatrix} -\tilde{Q}_{i,n} & \bar{A}_{i,m}\bar{G}_{i,m}+\bar{B}_{i,m}\bar{Y}_{i,m} \\ * & (1-\alpha_i)(\bar{Q}_{i,m}-\bar{G}_{i,m}^T-\bar{G}_{i,m}), \end{bmatrix} \leq 0 \tag{31}
$$

which is equivalent to inequality (B5), where $\tilde{Q}_{i,n} \triangleq (\bar{P}_{i,n}+\varepsilon I)^{-1} = \left(\begin{bmatrix} P_{i,n} & 0_{n\times 1} \\ * & 0 \end{bmatrix} + \varepsilon I\right)^{-1}$ and $\bar{Q}_{i,m} \triangleq \bar{P}_{i,m}^{-1}$, $\bar{Y}_{i,m} \triangleq \bar{K}_{i,m}\bar{G}_{i,m}$. Moreover, at the switching time instant, that is, $x_{k+1} \in \digamma_{j,n}$, $x_k \in \digamma_{i,m}$, $\forall (m,n) \in \tilde{\mathcal{S}}$, $m \in \mathcal{S}_{i,1}$, $n \in \mathcal{S}_{j,0}$, and $i \neq j$, the following inequalities need to be assured with the aid of Lemma 2:

$$
\tilde{P}_{j,n} \leq \mu_i \bar{P}_{i,m},\ i \neq j,
$$

where $\tilde{P}_{j,n} \triangleq \bar{P}_{j,n} + \varepsilon I$ with $\bar{P}_{j,n} \triangleq \begin{bmatrix} P_{j,n} & 0_{n\times 1} \\ * & 0 \end{bmatrix}$, and $\varepsilon < 10^{-5}$. Then one has,

$$
\bar{Q}_{i,m} \leq \mu_i \tilde{Q}_{j,n},\ i \neq j \tag{32}
$$

which is equivalent to the inequality (B6), where $\tilde{Q}_{j,n} \triangleq \tilde{P}_{j,n}^{-1}$ and $\bar{Q}_{i,m} \triangleq \bar{P}_{i,m}^{-1}$.

Case IV: Assuming that $x_{k+1} \in \digamma_{i,n}$, $x_k \in \digamma_{i,m}$ and $\forall (m,n) \in$

$\bar{\mathcal{S}}$, $m \in \mathcal{S}_{i,0}$, $n \in \mathcal{S}_{i,1}$ for the same PWA subsystem i, we have

$$\begin{aligned}
&\Delta V_i(\bar{x}_{k+1}, x_k, m) \\
=\;& \bar{x}_{k+1}^T \bar{P}_{i,n} \bar{x}_{k+1} - x_k^T P_{i,m} x_k \\
=\;& \bar{x}_k^T \breve{A}_{i,m}^T \bar{P}_{i,n} \breve{A}_{i,m} \bar{x}_k - x_k^T P_{i,m} x_k \\
=\;& \bar{x}_k^T \left\{ \breve{A}_{i,m}^T \bar{P}_{i,n} \breve{A}_{i,m} - (1-\alpha_i) \begin{bmatrix} P_{i,m} & 0_{n\times 1} \\ * & 0 \end{bmatrix} \right\} \bar{x}_k \\
& -\alpha_i V(x_k) \\
=\;& \bar{x}_k^T M_4 \bar{x}_k - \alpha_i V(x_k),
\end{aligned}$$

where $\breve{A}_{i,m} \triangleq \begin{bmatrix} A_{i,m} + B_{i,m} K_{i,m} & 0 \\ 0 & 0 \end{bmatrix}$, $M_4 \triangleq (\bar{A}_{i,m} + \bar{B}_{i,m}\bar{K}_{i,m})^T \bar{P}_{i,n} (\bar{A}_{i,m} + \bar{B}_{i,m}\bar{K}_{i,m}) - (1-\alpha_i)\bar{P}_{i,m}$, $\bar{A}_{i,m} \triangleq \begin{bmatrix} A_{i,m} & 0 \\ 0 & 1 \end{bmatrix}$, $\bar{K}_{i,m} \triangleq \begin{bmatrix} K_{i,m} & 0 \end{bmatrix}$, and $\bar{P}_{i,m}$ is irreversible. If $M_4 \le 0$, one has that inequality (12) holds. Applying Schur supplement to M_4, similar to the *Case III*, there holds that

$$\begin{bmatrix} -\bar{Q}_{i,n} & \bar{A}_{i,m}\tilde{G}_{i,m} + \bar{B}_{i,m}\tilde{Y}_{i,m} \\ * & (1-\alpha_i)(\tilde{Q}_{i,m} - \tilde{G}_{i,m}^T - \tilde{G}_{i,m}) \end{bmatrix} \le 0, \tag{33}$$

which implies that inequality (B7) holds, where $\tilde{Q}_{i,m} \triangleq \left(\begin{bmatrix} P_{i,m} & 0_{n\times 1} \\ * & 0 \end{bmatrix} + \varepsilon I \right)^{-1}$, $\bar{Q}_{i,n} \triangleq \bar{P}_{i,n}^{-1}$, and $\tilde{Y}_{i,m} \triangleq \bar{K}_{i,m}\tilde{G}_{i,m}$. Furthermore, at the switching time instant, that is, $x_{k+1} \in \digamma_{j,n}$, $x_k \in \digamma_{i,m}$, $\forall (m,n) \in \tilde{\mathcal{S}}$, $m \in \mathcal{S}_{i,0}$, $n \in \mathcal{S}_{j,1}$, and $i \neq j$, by (20), the inequality $\bar{P}_{j,n} \le \mu_i \tilde{P}_{i,m}$, $i \neq j$ needs to be satisfied, where $\tilde{P}_{i,m} \triangleq \bar{P}_{i,m} + \varepsilon I$ with $\bar{P}_{i,m} \triangleq \begin{bmatrix} P_{i,m} & 0_{n\times 1} \\ * & 0 \end{bmatrix}$ and $\varepsilon < 10^{-5}$. Then it can be directly derived that $\tilde{Q}_{i,m} \le \mu_i \bar{Q}_{j,n}$, $i \neq j$, which is equivalent to inequality (B8), where $\tilde{Q}_{i,m} \triangleq \tilde{P}_{i,m}^{-1}$ and $\bar{Q}_{j,n} \triangleq \bar{P}_{j,n}^{-1}$.

Therefore, based on the analyses of *Cases I-IV*, it can be concluded from Lemma 2 that the closed-loop switched PWA systems (24) is GUAS, and the state-feedback controller gains can be determined by $\bar{K}_{i,m} = \bar{Y}_{i,m}\bar{G}_{i,m}^{-1}$ and $K_{i,m} = Y_{i,m}G_{i,m}^{-1}$. The proof is complete. ■

Remark 5: In Theorem 2, the extra degree of freedom introduced by the matrix $\bar{G}_{i,m}$ $(G_{i,m})$, which is not considered symmetric, is incorporated in the control variable. Hence, it is clearly seen that the determination of the control law does not depend explicitly on the Lyapunov matrices $\bar{P}_{i,m}$ $(P_{i,m})$ by means of this change of variables. The multiple Lyapunov functions in the PWA form are used here to perform the PWA state-feedback controller design with the aid of Lemma 2. As an extension, a lifting process [20], ensuring the convexification of conditions, can also be borrowed here to deal with the l_2-gain analysis and $\mathcal{H}_\infty$ control synthesis issue for discrete-time switched PWA systems with dwell-time switching.

V. Numerical Example

In this section, one numerical example is presented to demonstrate the potential and validity of our developed theoretical results. Our purpose here is to check the GUAS of the considered nominal switched PWA systems and to design a stabilizing PWA controller aiming to the closed-loop switched PWA systems. Note that a potential application of our proposed approaches will be the pneumatic artificial muscles (PAMs) that were concerned in [30], where the PAMs dynamics during inflation and deflation is approximated by a class of switched PWA systems.

Consider the discrete-time switched PWA system (4) with PWA system 1:

$$A_{1,1} = \begin{bmatrix} -0.0600 & -0.6915 \\ -0.2085 & 0.5115 \end{bmatrix}, \quad B_{1,1} = \begin{bmatrix} -0.5 \\ 0.8 \end{bmatrix},$$

$$A_{1,2} = \begin{bmatrix} 1.4040 & 0.4845 \\ 1.1820 & -0.0735 \end{bmatrix}, \quad B_{1,2} = \begin{bmatrix} 0.3 \\ 0.9 \end{bmatrix},$$

TABLE I: Values of $\tau_{d,i}^*, i = 1, 2$ for given different pairs of (α_1, α_2).

α_i	$\tau_{d,i}^*$
$\alpha_1 = 0.4,\ \alpha_2 = 0.6$	$\tau_{d,1}^* = 3.0448,\ \tau_{d,2}^* = 4.2107$
$\alpha_1 = 0.3,\ \alpha_2 = 0.6$	$\tau_{d,1}^* = 4.9201,\ \tau_{d,2}^* = 4.1559$
$\alpha_1 = 0.2,\ \alpha_2 = 0.5$	$\tau_{d,1}^* = 9.3742,\ \tau_{d,2}^* = 6.9459$
$\alpha_1 = 0.1,\ \alpha_2 = 0.4$	$\tau_{d,1}^* = 25.2335,\ \tau_{d,2}^* = 10.2535$

$$A_{1,3} = \begin{bmatrix} -1.2855 & 1.2225 \\ 0.7365 & 0.9300 \end{bmatrix}, \quad B_{1,3} = \begin{bmatrix} -0.6 \\ 0.4 \end{bmatrix},$$

$$A_{1,4} = \begin{bmatrix} -0.0330 & 0.9660 \\ 1.1370 & 0.4065 \end{bmatrix}, \quad B_{1,4} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

$$H_{1,1} = \mathrm{diag}\{1, 1\},\ H_{1,2} = \mathrm{diag}\{1, -1\},\ H_{1,3} = \mathrm{diag}\{-1, -1\},$$

$$H_{1,4} = \mathrm{diag}\{-1, 1\},\ h_{1,m}, a_{1,m} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},\ m = 1, 2, 3, 4;$$

PWA system 2:

$$A_{2,1} = \begin{bmatrix} -1.8301 & 0.1590 \\ 1.8000 & 0.5768 \end{bmatrix}, \quad B_{2,1} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix},$$

$$H_{2,1} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}, \quad h_{2,1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad a_{2,1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$A_{2,2} = \begin{bmatrix} -0.7588 & -0.9125 \\ 0.6777 & 1.6991 \end{bmatrix}, \quad B_{2,2} = \begin{bmatrix} 0.2 \\ -0.3 \end{bmatrix},$$

$$H_{2,2} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad h_{2,2} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad a_{2,2} = \begin{bmatrix} 0 \\ 0.3 \end{bmatrix},$$

$$A_{2,3} = \begin{bmatrix} 0.3126 & 0.1590 \\ -0.4446 & 0.5768 \end{bmatrix}, \quad B_{2,3} = \begin{bmatrix} 0.4 \\ 0.1 \end{bmatrix},$$

$$H_{2,3} = \begin{bmatrix} -1 & 0 \\ -1 & -1 \end{bmatrix}, \quad h_{2,3} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad a_{2,3} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Obviously, we get that $\mathcal{S}_{1,0} = \{1, 2, 3, 4\}$, $\mathcal{S}_{1,1} = \varnothing$, $\mathcal{S}_{2,0} = \{1, 3\}$, $\mathcal{S}_{2,1} = \{2\}$. According to the reachability analysis approach, the sets $\tilde{\mathcal{S}}$ and $\bar{\mathcal{S}}$ can be determined, respectively, which collect the indices of reachable state partitions. For instance, it yields that $\bar{\mathcal{S}} = \{2, 3, 4\}$ for the condition (B6) in Theorem 2. Considering different values of the scalars α_1 and α_2, by Theorem 1 and some simple computations, we can determine the minimal MDT $\tau_{d,1}^*$ and $\tau_{d,2}^*$, respectively, as listed in Table I.

Designating that $\alpha_1 = 0.3$, $\alpha_2 = 0.6$, through solving (B1)–(B8) in Theorem 2, one can calculate the controller gain matrices as follows

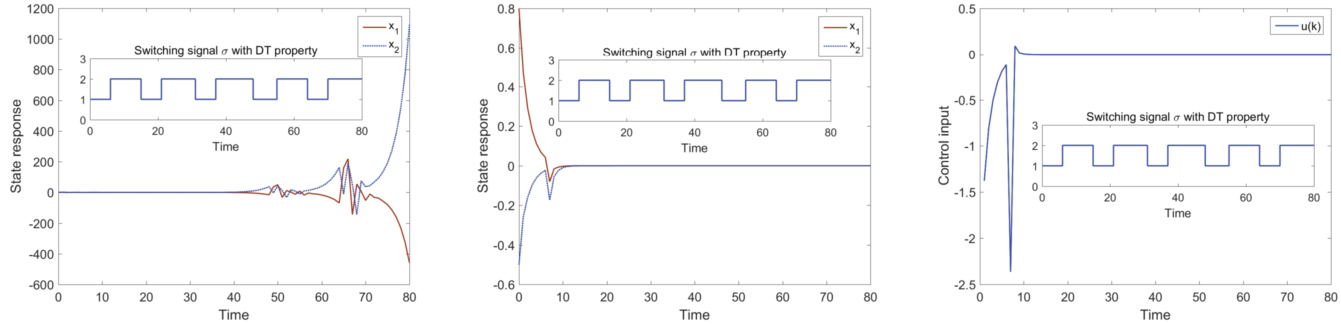
$$\begin{aligned}
&K_{1,1} = \begin{bmatrix} 0.1633 & -0.8300 \end{bmatrix},\ K_{1,3} = \begin{bmatrix} -2.0272 & 0.3762 \end{bmatrix}, \\
&K_{1,2} = \begin{bmatrix} -2.0272 & 0.3762 \end{bmatrix},\ K_{1,4} = \begin{bmatrix} 0.0369 & -0.9633 \end{bmatrix}, \\
&K_{2,1} = \begin{bmatrix} -56.5728 & -0.7424 \end{bmatrix},\ K_{2,3} = \begin{bmatrix} -1.1778 & 0.0095 \end{bmatrix}, \\
&K_{2,2} = \begin{bmatrix} 2.7313 & 5.3248 & 0.6923 \end{bmatrix}.
\end{aligned}$$

Furthermore, under the initial condition $x_0 = [0.8; -0.5]$, using the Multi-Parametric Toolbox [27], the open-loop and closed-loop state responses, as well as the control input are plotted in Fig. 2. From Fig. 2, it can be clearly seen that the developed PWA state-feedback controller can stabilize the closed-loop system (24) even though the state trajectories of the open-loop system are not convergent eventually.

VI. Conclusion

The stability analysis and state-feedback control design issues have been studied for a class of discrete-time switched nonlinear systems via the smooth approximation method. The dual switching mechanism including both modal dwell-time switching and state-partition-dependent switching has been considered for the first time to

This article has been accepted for publication in a future issue of this journal, but has not been fully edited. Content may change prior to final publication. Citation information: DOI 10.1109/TAC.2019.2938302, IEEE Transactions on Automatic Control



(a) (b) (c)

Fig. 2: (a) The open-loop state response. (b) The closed-loop state response. (c) The control input.

regulate the switchings among nonlinear subsystems and within each subsystem. Together with the state partition knowledge, the multiple Lyapunov functions in piecewise quadratic form have been exploited to derive the sufficient conditions ensuring the asymptotic stability with less conservatism. The existence of piecewise affine controller gains matrices has been examined for the approximated switched PWA systems. The numerical example presented has verified that the developed theoretical results are feasible and effective. A possible extension in the future work will be to conduct the l_2-gain analysis and $\mathcal{H}_\infty$ control synthesis for discrete-time switched PWA systems through the convex lifting approach, where the autonomous switching sequence depends on both the state and input trajectories.

REFERENCES

[1] Z. Sun and S. S. Ge, *Stability Theory of Switched Dynamical Systems*. London: Springer-Verlag, 2011.

[2] Y. Wang, X. Sun, and F. Mazenc, “Stability of switched nonlinear systems with delay and disturbance,” *Automatica*, vol. 69, pp. 78–86, 2016.

[3] Z. Sun, “A note on marginal stability of switched systems,” *IEEE Transactions on Automatic Control*, vol. 53, no. 2, pp. 625–631, 2008.

[4] L. Long and J. Zhao, “An integral-type multiple Lyapunov functions approach for switched nonlinear systems,” *IEEE Transactions on Automatic Control*, vol. 61, no. 7, pp. 1979–1986, 2016.

[5] M. Hajiahmadi, B. De Schutter, and H. Hellendoorn, “Design of stabilizing switching laws for mixed switched affine systems,” *IEEE Transactions on Automatic Control*, vol. 61, no. 6, pp. 1676–1681, 2016.

[6] Y. Zhu, Z. Zhong, M. V. Basin, and D. Zhou, “A descriptor system approach to stability and stabilization of discrete-time switched PWA systems,” *IEEE Transactions on Automatic Control*, vol. 63, no. 10, pp. 3456–3463, 2018.

[7] S. Lim and J. P. How, “Modeling and $\mathcal{H}_\infty$ control for switched linear parameter-varying missile autopilot,” *IEEE Transactions on Control Systems Technology*, vol. 11, no. 6, pp. 830–838, 2003.

[8] L. Zhang, S. Zhuang, and R. D. Braatz, “Switched model predictive control of switched linear systems: Feasibility, stability and robustness,” *Automatica*, vol. 67, pp. 8–21, 2016.

[9] D. Mignone, G. Ferrari-Trecate, and M. Morari, “Stability and stabilization of piecewise affine and hybrid systems: An LMI approach,” in *Proceedings of the 39th IEEE Conference on Decision and Control*, Sydney, Australia, December 2000, pp. 504–509.

[10] G. Feng, “Stability analysis of piecewise discrete-time linear systems,” *IEEE Transactions on Automatic Control*, vol. 47, no. 7, pp. 1108–1112, 2002.

[11] L. Zhang, Z. Ning, and W. X. Zheng, “Observer-based control for piecewise-affine systems with both input and output quantization,” *IEEE Transactions on Automatic Control*, vol. 62, no. 11, pp. 5858–5865, 2017.

[12] F. A. Cuzzola and M. Morari, “An LMI approach for $\mathcal{H}_\infty$ analysis and control of discrete-time piecewise affine systems,” *International Journal of Control*, vol. 75, no. 16–17, pp. 1293–1301, 2002.

[13] A. S. Morse, “Supervisory control of families of linear set-point controllers-Part I: Exact matching,” *IEEE Transactions on Automatic Control*, vol. 41, no. 10, pp. 1413–1431, 1996.

[14] A. Mendoza-Torres, N. Visairo, C. Nuñez, J. Armenta, E. Rodríguez, and I. Cervantes, “Switching rule for a bidirectional DC/DC converter in an electric vehicle,” *Control Engineering Practice*, vol. 82, pp. 108–117, 2019.

[15] R. Morselli, R. Zanasi, and P. Ferracin, “Modelling and simulation of static and coulomb friction in a class of automotive systems,” *International Journal of Control*, vol. 79, no. 5, pp. 508–520, 2006.

[16] H. Ishii and B. A. Francis, “Stabilizing a linear system by switching control with dwell time,” *IEEE Transactions on Automatic Control*, vol. 47, no. 12, pp. 1962–1973, 2002.

[17] W.-H. Chen and W. X. Zheng, “Delay-independent minimum dwell time for exponential stability of uncertain switched delay systems,” *IEEE Transactions on Automatic Control*, vol. 55, no. 10, pp. 2406–2413, 2010.

[18] J. C. Geromel and P. Colaneri, “Stability and stabilization of discrete-time switched systems,” *International Journal of Control*, vol. 79, no. 07, pp. 719–728, 2006.

[19] X. Zhao, P. Shi, Y. Yin, and S. K. Nguang, “New results on stability of slowly switched systems: A multiple discontinuous Lyapunov function approach,” *IEEE Transactions on Automatic Control*, vol. 62, no. 7, pp. 3502–3509, 2017.

[20] C. Briat, “Convex lifted conditions for robust l_2-stability analysis and l_2-stabilization of linear discrete-time switched systems with minimum dwell-time constraint,” *Automatica*, vol. 50, no. 3, pp. 976–983, 2014.

[21] W. Xiang and J. Xiao, “Convex sufficient conditions on asymptotic stability and l_2-gain performance for uncertain discrete-time switched linear systems,” *IET Control Theory & Applications*, vol. 8, no. 3, pp. 211–218, 2014.

[22] G. Chesi and P. Colaneri, “Homogeneous rational Lyapunov functions for performance analysis of switched systems with arbitrary switching and dwell time constraints,” *IEEE Transactions on Automatic Control*, vol. 62, no. 10, pp. 5124–5137, 2017.

[23] W. Xiang, H. D. Tran, and T. T. Johnson, “Nonconservative lifted convex conditions for stability of discrete-time switched systems under minimum dwell-time constraint,” *IEEE Transactions on Automatic Control*, vol. 64, no. 8, pp. 3407–3414, 2019.

[24] M. S. Branicky, “Multiple Lyapunov functions and other analysis tools for switched and hybrid systems,” *IEEE Transactions on Automatic Control*, vol. 43, no. 4, pp. 475–482, 1998.

[25] J. C. Geromel and P. Colaneri, “Stability and stabilization of continuous-time switched linear systems,” *SIAM Journal on Control and Optimization*, vol. 45, no. 5, pp. 1915–1930, 2006.

[26] Y. Chen, S. Fei, K. Zhang, and Z. Fu, “Control synthesis of discrete-time switched linear systems with input saturation based on minimum dwell time approach,” *Circuits, Systems, and Signal Processing*, vol. 31, no. 2, pp. 779–795, 2012.

[27] M. Kvasnica, P. Grieder, and M. Baotić, Multi-Parametric Toolbox (MPT), 2006.

[28] M. C. de Oliveira, J. Bernussou, and J. C. Geromel, “A new discrete-time robust stability condition,” *Systems & Control Letters*, vol. 37, no. 4, pp. 261–265, 1999.

[29] S. Prajna and A. Papachristodoulou, “Analysis of switched and hybrid systems-beyond piecewise quadratic methods,” in *Proceedings of the 42nd IEEE Conference on Decision and Control*, Maui, Hawaii, USA, December 2003, pp. 2779–2784.

[30] G. Andrikopoulos, G. Nikolakopoulos, and S. Manesis, “Pneumatic artificial muscles: A switching model predictive control approach,” *Control Engineering Practice*, vol. 21, no. 12, pp. 1653–1664, 2013.